Name

Mathematics Extension 1 Year 12

Proof by Mathematical Induction

Book 1

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Syllabus Content

ME1-12-01 uses mathematical induction to prove results involving sums and divisibility

- Examine the nature of inductive proof, including the statement to be proved, the base case and the inductive step
- Prove results for sums using mathematical induction
- Prove divisibility results using mathematical induction
- Identify errors in false ‘proofs by induction’, such as cases where only one of the two required steps of a proof by induction is true

Proofs of Sums

Investigation The nature of proof.

In mathematical proofs, we start with a claim that can be either true or false, also known as a **proposition.**

Proposition: $n^2 - n + 11$ is prime for all natural numbers $n \in \{1, 2, 3, 4 \ldots\}$.

Let's start by checking if it works:

$n = 1,$ ……

$n = 2,$ ……

$n = 3,$ ……

$n = 4,$ ……

$n = 5,$ ……

$n = 6,$ ……

At first glance, the rule seems to work!

To **prove** whether a proposition is true for false:

- True if it works for all values of n.
- False if there is at least 1 single counterexample.

In mathematical proof, a single counterexample is enough to disprove something.

Find a counterexample.

Key Ideas

1. To prove a proposition is true, we must prove it is true for …… cases.
2. To prove a proposition is false, we can find a ……

Investigation Sum of whole numbers.

a Consider the sum: $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$

What is the result?

b Consider the sum: $1 + 2 + 3 + \cdots + 98 + 99 + 100$

Pairs add up to: ……

Number of pairs: ……

Total: ……

c Consider the sum $1 + 2 + 3 + \cdots + 998 + 999 + 1000$

Pairs add up to: ……

Number of pairs: ……

Total: ……

d Consider the sum $1 + 2 + 3 + \cdots + (n-2) + (n-1) + n$

Pairs add up to: ……

Number of pairs: ……

Total: ……

We have come up with a formula for summing any number of whole numbers

$$1 + 2 + 3 + \cdots + n = \frac{n}{2}(n + 1)$$

But how can we know for sure this works for *any* number?

We use a method called proof by **mathematical induction**

Here are the steps:

1. Prove that the base case is true, $n = 1$

For $n = 1$:

$LHS =$ $RHS =$

2. Assume that the kth case is true, $k \geq 1$ and k is a natural number.

$1 + 2 + 3 + \cdots + k =$ .....

3. Prove that the $(k + 1)^{\text{th}}$ case is true, using the assumption.

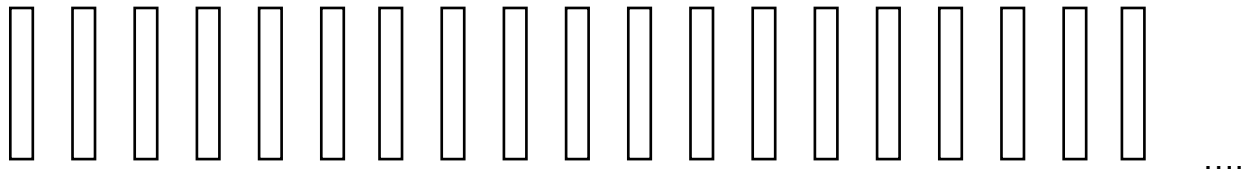
WTP: $1 + 2 + 3 + \cdots + k + (k + 1) =$ .....

$LHS =$

4. State proof by induction.

ⓘ Proof by Mathematical Induction

Imagine you have in infinite line of dominoes.

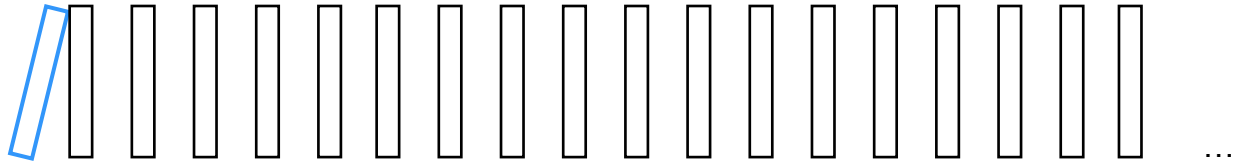


Each domino represents a proposition.

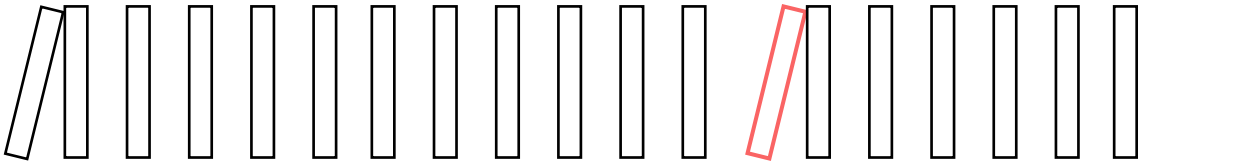
If a domino falls, then the proposition is true.

We want to prove that all dominoes will fall over.

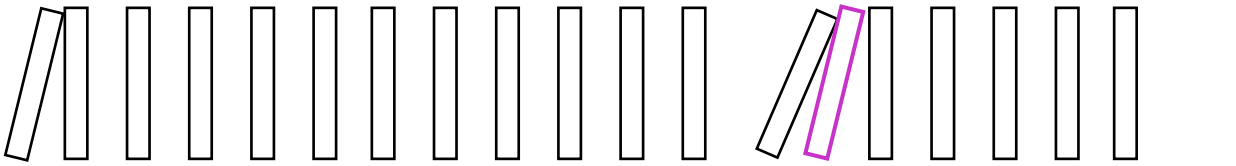
Step 1: Test that the first domino falls.



Step 2: Assume that the kth domino will fall.



Step 3: Prove that the $(k+1)^{\text{th}}$ domino will always fall.

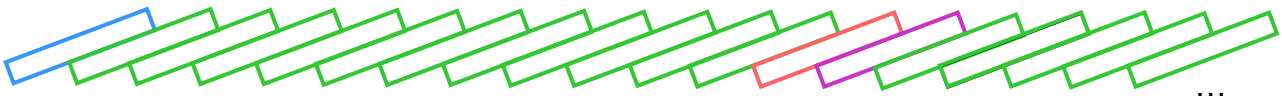


Step 4: Conclusion

If any domino falls, the next will always fall.

The first domino falls, so every domino will fall.

Therefore, proven by mathematical induction.



ⓘ Proof by Mathematical Induction

We can use mathematical induction to prove statements where n is a positive whole number.

1. Test the base case, usually $n = 1$
2. Assume $n = k$ is true.
3. Prove $n = k + 1$ is true.
4. State proof by induction.

⚠ You *must* set out your work in the above way to give a coherent proof.

Example Prove sums by mathematical induction.

Prove by mathematical induction:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 \quad \text{for all } n = 1,2,3 \ldots$$

1. Test $n = 1$

$LHS = 1$ $\qquad$ $RHS = 1^2 = 1$

$LHS = RHS$

$\therefore$ True for $n = 1$

2. Assume true for $n = k$

$1 + 3 + 5 + \cdots (2k - 1) = k^2$ for some positive integer k.

3. Prove $n = k + 1$

WTP: $1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1)) = (k + 1)^2$

$1 + 3 + 5 + \cdots (2k - 1) + (2k + 1) = (k + 1)^2$

$LHS = \underline{1 + 3 + 5 + \cdots + (2k - 1)} + (2k + 1)$

$= \underline{k^2} + (2k + 1)$ using assumption

$= k^2 + 2k + 1$

$= (k + 1)^2$

$= RHS$

4. State proof by MI

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

Practice

Prove by mathematical induction:

$$5 + 9 + 13 + \cdots + (4n + 1) = n(2n + 3) \quad \text{for all } n = 1,2,3 \ldots$$

1. Test $n = 1$

2. Assume true for $n = k$

3. Prove $n = k + 1$

4. State proof by MI

Prove by mathematical induction:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n + 1)(2n + 1) \quad \text{for } n = 1,2,3 \ldots$$

1 Prove by mathematical induction.

a

$1 + 2 + 2^2 + \cdots + 2^{n-1} = 2^n - 1 \quad \text{for all } n \geq 1$

1. Test $n = 1$

$LHS = 1 \qquad RHS = 2^1 - 1 = 1$

$LHS = RHS$

$\therefore$ True for $n = 1$

2. Assume true for $n = k$

$1 + 2 + 2^2 + \cdots + 2^{k-1} = 2^k - 1$

3. Prove $n = k + 1$

WTP: $1 + 2 + 2^2 + \cdots + 2^{k-1} + 2^{k+1-1} = 2^{k+1} - 1$

$LHS = 1 + 2 + 2^2 + \cdots + 2^{k-1} + 2^{k+1-1}$

$= 2^k - 1 + 2^{k+1-1}$, using assumption

$= 2^k + 2^k - 1$

$= 2 \times 2^k - 1$

$= 2^{k+1} - 1$

$= RHS$

4. State proof by MI

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

b

$2 + 4 + 8 + \ldots + 2^n = 2(2^n - 1) \quad \text{for all } n \geq 1$

Test $n = 1$

$LHS = 2 \qquad RHS = 2(2^1 - 1) = 2$

$LHS = RHS$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$2 + 4 + 8 + \cdots + 2^k = 2(2^k - 1)$

Prove $n = k + 1$

WTP: $2 + 4 + 8 + \cdots + 2^k + 2^{k+1} = 2(2^{k+1} - 1)$

$LHS = [2 + 4 + 8 + \cdots + 2^k] + 2^{k+1}$

$= 2(2^k - 1) + 2^{k+1}$, using assumption

$= 2\hat{}(k+1) - 2 + 2^{k+1}$

$= 2 \times 2^{k+1} - 2$

$= 2(2^{k+1} - 1)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

c

$$\frac{1}{1(2)}+\frac{1}{2(3)}+\frac{1}{3(4)}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1} \quad \text{for all } n \geq 1$$

Test $n = 1$:

$$LHS = \frac{1}{1\times 2} \qquad RHS = \frac{1}{1+1}$$
$$= \frac{1}{2} \qquad = \frac{1}{2}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$$\frac{1}{1(2)}+\frac{1}{2(3)}+\frac{1}{3(4)}+\cdots+\left(\frac{1}{k(k+1)}\right)=\frac{k}{k+1}$$

Prove $n = k + 1$:

$$WTP: \frac{1}{1(2)}+\frac{1}{2(3)}+\frac{1}{3(4)}+\cdots+\left(\frac{1}{k(k+1)}\right)+\left(\frac{1}{(k+1)\big((k+1)+1\big)}\right)=\frac{k+1}{(k+1)+1}$$

$$LHS = \frac{1}{1(2)}+\frac{1}{2(3)}+\frac{1}{3(4)}+\cdots+\left(\frac{1}{k(k+1)}\right)+\left(\frac{1}{(k+1)\big((k+1)+1\big)}\right)$$
$$= \frac{k}{k+1}+\left(\frac{1}{(k+1)\big((k+1)+1\big)}\right), \text{using assumption}$$
$$= \frac{k}{k+1}+\left(\frac{1}{(k+1)(k+2)}\right)$$
$$= \frac{k(k+2)}{(k+1)(k+2)}+\frac{1}{(k+1)(k+2)}$$
$$= \frac{k^2+2k}{(k+1)(k+2)}+\frac{1}{(k+1)(k+2)}$$
$$= \frac{k^2+2k+1}{(k+1)(k+2)}$$
$$= \frac{(k+1)^2}{(k+1)(k+2)}$$
$$= \frac{k+1}{k+2}$$
$$= \frac{k+1}{(k+1)+1}$$
$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

d

$$\frac{1}{1(3)}+\frac{1}{3(5)}+\frac{1}{5(7)}+\cdots+\frac{1}{(2n-1)(2n+1)}=\frac{n}{2n+1} \quad \text{for all } n \geq 1$$

Test $n = 1$:

$$LHS = \frac{1}{1\times 3} \qquad RHS = \frac{1}{2(1)+1}$$
$$= \frac{1}{3} \qquad = \frac{1}{3}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$$\frac{1}{1(3)}+\frac{1}{3(5)}+\frac{1}{5(7)}+\cdots+\left(\frac{1}{(2k-1)(2k+1)}\right)=\frac{k}{2k+1}$$

Prove $n = k + 1$:

$$WTP: \frac{1}{1(3)}+\frac{1}{3(5)}+\frac{1}{5(7)}+\cdots+\left(\frac{1}{(2k-1)(2k+1)}\right)+\left(\frac{1}{(2(k+1)-1)(2(k+1)+1)}\right)=\frac{k+1}{2(k+1)+1}$$

$$LHS = \frac{1}{1(3)}+\frac{1}{3(5)}+\frac{1}{5(7)}+\cdots+\left(\frac{1}{(2k-1)(2k+1)}\right)+\left(\frac{1}{(2(k+1)-1)(2(k+1)+1)}\right)$$
$$= \frac{k}{2k+1}+\left(\frac{1}{(2(k+1)-1)(2(k+1)+1)}\right), \text{ using assumption}$$
$$= k/(2k+1)+\left(\frac{1}{(2k+1)(2k+3)}\right)$$
$$= \frac{k(2k+3)}{(2k+1)(2k+3)}+\frac{1}{(2k+1)(2k+3)}$$
$$= \frac{2k^2+3k}{(2k+1)(2k+3)}+\frac{1}{(2k+1)(2k+3)}$$
$$= \frac{2k^2+3k+1}{(2k+1)(2k+3)}$$
$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$
$$= \frac{k+1}{2k+3}$$
$$= \frac{k+1}{2(k+1)+1}$$
$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

e

$5 + 10 + 15 + \ldots + 5n = \frac{5}{2}n(n + 1) \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 5 \qquad RHS = \frac{5(1)(1 + 1)}{2} = 5$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$5 + 10 + 15 + \cdots + 5k = \frac{5k(k + 1)}{2}$

Prove $n = k + 1$:

$WTP: 5 + 10 + 15 + \cdots + 5k + [5(k + 1)] = \frac{5(k + 1)(k + 2)}{2}$

$LHS = [5 + 10 + 15 + \cdots + 5k] + \big(5(k + 1)\big)$

$= \frac{5k(k + 1)}{2} + 5(k + 1), \text{ using assumption}$

$= \frac{5k(k + 1) + 10(k + 1)}{2}$

$= \frac{5(k + 1)(k + 2)}{2}$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

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f

$-2 - 4 - 6 - \ldots - 2n = -n(n + 1) \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = -2 \qquad RHS = -1(1 + 1) = -2$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$-2 - 4 - 6 - \cdots - 2k = -k(k + 1)$

Prove $n = k + 1$:

$WTP: -2 - 4 - 6 - \cdots - 2k - \big(2(k + 1)\big) = -(k + 1)(k + 2)$

$LHS = [-2 - 4 - 6 - \cdots - 2k] - [2(k + 1)]$

$= -k(k + 1) - [2(k + 1)], \text{using assumption}$

$= (k + 1)(-k - 2)$

$= -(k + 1)(k + 2)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

g

$1 + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^{n-1}} = 2\left(1 - \frac{1}{2^n}\right)$ for all $n \geq 1$

Test $n = 1$

$LHS = 1$ $\quad RHS = 2\left(1 - \frac{1}{2^1}\right) = 1$

$\therefore$ True for $n = 1$

Assume $n = k$

$1 + \left(\frac{1}{2}\right) + \left(\frac{1}{4}\right) + \cdots + \left(\frac{1}{2^{k-1}}\right) = 2\left(1 - \frac{1}{2^k}\right)$

Prove $n = k + 1$

$WTP: 1 + \left(\frac{1}{2}\right) + \left(\frac{1}{4}\right) + \cdots + \left(\frac{1}{2^{k-1}}\right) + \left(\frac{1}{2^k}\right) = 2\left(1 - \left(\frac{1}{2^{k+1}}\right)\right)$

$LHS = \left[1 + \left(\frac{1}{2}\right) + \left(\frac{1}{4}\right) + \cdots + \left(\frac{1}{2^{k-1}}\right)\right] + \left(\frac{1}{2^k}\right)$

$= 2\left(1 - \frac{1}{2^k}\right) + \left(\frac{1}{2^k}\right)$, using assumption

$= 2 - \frac{2}{2^k} + \frac{1}{2^k}$

$= 2 - \frac{1}{2^k}$

$= 2\left(1 - \frac{1}{2^{k+1}}\right)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

h

$1 + 5 + 5^2 + \cdots + 5^{n-1} = \frac{1}{4}(5^n - 1) \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 1 \qquad RHS = \frac{1}{4}(5^1 - 1)$

$= 1$

$LHS = RHS$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$1 + 5 + 5^2 + \cdots + 5^{k-1} = \frac{1}{4}(5^k - 1)$

Prove true for $n = k + 1$:

WTP: $1 + 5 + 5^2 + \cdots + 5^{k-1} + 5^{k+1-1} = \frac{1}{4}(5^{k+1} - 1)$

$\text{LHS} = 1 + 5 + 5^2 + \cdots + 5^{k-1} + 5^{k+1-1}$

$= \frac{1}{4}(5^k - 1) + 5^{k+1-1}, \text{using assumption}$

$= \frac{1}{4}(5^k - 1 + 4 \times 5^{k+1-1})$

$= \frac{1}{4}(5^k - 1 + 4 \times 5^k)$

$= \frac{1}{4}(5 \times 5^k - 1)$

$= \frac{1}{4}(5^{k+1} - 1)$

$= \text{RHS}$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

i

$1(1!) + 2(2!) + 3(3!) + \cdots + n(n!) = (n+1)! - 1 \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 1 \times 1! = 1 \qquad RHS = (1+1)! - 1 = 2 - 1 = 1$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$1(1!) + 2(2!) + 3(3!) + \cdots + k(k!) = (k+1)! - 1$

Prove $n = k + 1$:

$WTP: 1(1!) + 2(2!) + 3(3!) + \cdots + k(k!) + (k+1)(k+1)! = (k+2)! - 1$

$$\begin{aligned} LHS &= 11(1!) + 2(2!) + 3(3!) + \cdots + k(k!) + (k+1)(k+1)! \\ &= (k+1)! - 1 + (k+1)(k+1)!, \text{using assumption} \\ &= (k+1)! + (k+1)(k+1)! - 1 \\ &= (k+1)!(1 + k + 1) - 1 \\ &= (k+1)!(k+2) - 1 \\ &= (k+2)! - 1 \\ &= RHS \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

j

$2 + 5 + 8 + \ldots + (3n - 1) = \dfrac{3n^2 + n}{2} \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 2 \qquad RHS = \dfrac{3(1^2) + 1}{2} = 2$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$2 + 5 + 8 + \cdots + (3k - 1) = \dfrac{3k^2 + k}{2}$

Prove $n = k + 1$:

$$WTP: 2 + 5 + 8 + \cdots + (3k - 1) + (3(k+1) - 1) = \frac{3(k+1)^2 + (k+1)}{2} = \frac{3k^2 + 7k + 4}{2}$$

$$\begin{aligned} LHS &= [2 + 5 + 8 + \cdots + (3k - 1)] + [3(k+1) - 1] \\ &= \frac{3k^2 + k}{2} + 3k + 2, \text{using assumption} \\ &= \frac{3k^2 + 7k + 4}{2} \\ &= RHS \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

k Suppose that $1 + 3 + 5 + \cdots + (2n - 1) = n^2 + 2$ is true for $n = k$

Prove that it is true for $n = k + 1$.

$WTP = 1 + 3 + 5 + \cdots (2k - 1) + (2(k + 1) - 1) = (k + 1)^2 + 2$

$LHS = 1 + 3 + 5 + \cdots (2k - 1) + (2(k + 1) - 1)$

$= k^2 + 2 + (2(k + 1) - 1)$, using assumption $1 + 3 + 5 + \cdots (2k - 1) = k^2 + 2$

$= k^2 + 2 + (2k + 2 - 1)$

$= k^2 + 2k + 3$

$= (k + 1)^2 + 2$

Explain why we cannot conclude it is also true for all integers $n \geq 1$.

The base case $n = 1$ is not true.

l Prove or disprove $3 + 6 + 9 + \cdots + 3n = n(n + 1) + 1$ for all positive integers $n \geq 1$.

Disprove by counterexample: case $n = 2$ is not true.

m Explain why you cannot prove $3 + 8 + 13 + \cdots + (5n - 2) = \frac{n(5n+1)}{2}$ for all real numbers using mathematical induction.

Mathematical induction only works for positive whole number values of n.

n Explain why you cannot prove $\sin n \cot n = \cos n$ using mathematical induction.

Mathematical induction only works for positive whole number values of n. n can be any angle.

2 Prove the following statements using mathematical induction.

a

$3 + 10 + 17 + \ldots + (7n - 4) = \frac{n}{2}(7n - 1)$ for all $n \geq 1$

Test $n = 1$

$LHS = 3$ $\quad RHS = \frac{1}{2}(7(1) - 1) = \frac{1}{2}(6) = 3$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$3 + 10 + 17 + \cdots + (7k - 4) = \frac{k}{2}(7k - 1)$

Prove $n = k + 1$

$WTP: 3 + 10 + 17 + \cdots + (7k - 4) + (7(k + 1) - 4) = \left(\frac{k + 1}{2}\right)(7(k + 1) - 1)$

$= \left(\frac{k + 1}{2}\right)(7k - 6)$

$LHS = [3 + 10 + 17 + \cdots + (7k - 4)] + (7k + 3)$

$= \frac{k}{2}(7k - 1) + (7k + 3)$, using assumption

$= \frac{1}{2}(7k^2 - k) + 7k + 3$

$= \frac{1}{2}[7k^2 - k + 14k + 6]$

$= \frac{1}{2}(7k^2 + 13k + 6)$

$= \frac{1}{2}(k + 1)(7k + 6)$

$= \left(\frac{k+1}{2}\right)(7k + 6)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

b

$5 + 10 + 20 + \ldots + 5(2^n) = 5(2^{n+1} - 1)$ for all $n \geq 0$ ← The base case is $n = 0$

Test $n = 0$

$LHS = 5 \times 2^0 = 5$ $\quad RHS = 5(2^{0+1} - 1) = 5(2^1 - 1) = 5$

$\therefore$ True for $n = 0$

Assume true for $n = k$

$5 + 10 + 20 + \cdots + 5 \times 2^k = 5(2^{k+1} - 1)$

Prove $n = k + 1$

$WTP: 5 + 10 + 20 + \cdots + (5 \times 2^k) + (5 \times 2^{k+1}) = 5(2^{k+2} - 1)$

$$\begin{aligned} LHS &= [5 + 10 + 20 + \cdots + (5 \times 2^k)] + (5 \times 2^{k+1}) \\ &= 5(2^{k+1} - 1) + (5 \times 2^{k+1}), \text{using assumption} \\ &= 10 \times 2^{k+1} - 5 \\ &= 5(2 \times 2^{k+1} - 1) \\ &= 5(2^{k+2} - 1) \\ &= RHS \end{aligned}$$

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

c

$2(2^0) + 3(2^1) + 4(2^2) + \cdots + (n+1)(2^{n-1}) = n(2^n)$ for all $n \geq 1$

Test $n = 1$

$LHS = 2 \times 2^0 = 2$ $\quad RHS = 1 \times 2^1 = 2$

$\therefore$ True for n = 1

Assume true for $n = k$

$2(2^0) + 3(2^1) + 4(2^2) + \cdots + (k+1) \times 2^{k-1} = k \times 2^k$

Prove $n = k + 1$

$WTP: 2(2^0) + 3(2^1) + 4(2^2) + \cdots + (k+1) \times 2^{k-1} + \big((k+1) + 1\big) \times 2^{k+1-1}$

$= (k+1) \times 2^{k+1}$

$$\begin{aligned} LHS &= 2(2^0) + 3(2^1) + 4(2^2) + \cdots + (k+1) \times 2^{k-1} + \big((k+1) + 1\big) \times 2^{k+1-1} \\ &= k \times 2^k + \big((k+1) + 1\big) \times 2^{k+1-1}, \text{using assumption} \\ &= k \times 2^k + ((k+1) + 1) \times 2^k \\ &= k \times 2^k + (k+2) \times 2^k \\ &= (k + k + 2) \times 2^k \\ &= (2k + 2) \times 2^k \\ &= (k+1) \times 2 \times 2^k \\ &= (k+1) \times 2^{k+1} \\ &= RHS \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

d

$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1) \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 1 \qquad RHS = \left(\frac{1}{6}\right)(1)(1+1)(2(1)+1) = \frac{1}{6} \times 1 \times 2 \times 3 = 1$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{1}{6}k(k+1)(2k+1)$

Prove $n = k + 1$:

$WTP: 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{1}{6}(k+1)\big((k+1)+1\big)(2(k+1)+1)$

$= \frac{1}{6}(k+1)(k+2)(2k+3)$

$LHS = 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2$

$= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$, using assumption

$= (k+1) \times \left(\frac{1}{6}k(2k+1) + (k+1)\right)$

$= \frac{1}{6}(k+1)(k(2k+1) + 6k + 6)$

$= \frac{1}{6}(k+1)(2k^2 + k + 6k + 6)$

$= \frac{1}{6}(k+1)(2k^2 + 7k + 6)$

$= \frac{1}{6}(k+1)(k+2)(2k+3)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

e

$1(2) + 2(3) + 3(4) + \cdots + n(n+1) = \frac{1}{3}n(n+1)(n+2) \quad \text{for all } n \geq 1$

Test $n = 1$:

$LHS = 1 \times 2 = 2 \qquad RHS = \left(\frac{1}{3}\right)(1)(1+1)(1+2) = \frac{1}{3} \times 2 \times 3 = 2$

$\therefore$ True for $n = 1$

Assume true for $n = k$:

$1(2) + 2(3) + 3(4) + \cdots + k(k+1) = \frac{1}{3}k(k+1)(k+2)$

Prove $n = k + 1$

$WTP: 1(2) + 2(3) + 3(4) + \cdots + k(k+1) + (k+1)\big((k+1)+1\big)$

$= \frac{1}{3}(k+1)\big((k+1)+1\big)\big((k+1)+2\big)$

$= \frac{1}{3}(k+1)(k+2)(k+3)$

$LHS = 1(2) + 2(3) + 3(4) + \cdots + k(k+1) + (k+1)\big((k+1)+1\big)$

$= \frac{1}{3}k(k+1)(k+2) + (k+1)\big((k+1)+1\big)$, using assumption

$= \frac{1}{3}k(k+1)(k+2) + (k+1)(k+2)$

$= (k+1)(k+2)\left(\frac{1}{3}k + 1\right)$

$= \frac{1}{3}(k+1)(k+2)(k+3)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

f

$9 + 14 + 19 + \ldots + (5n + 4) = \dfrac{5n^2 + 13n}{2}$ for all $n \geq 1$

Test $n = 1$

$LHS = 9$ $\quad RHS = \dfrac{5(1^2) + 13(1)}{2} = 9$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$9 + 14 + 19 + \cdots + (5k + 4) = \dfrac{5k^2 + 13k}{2}$

Prove $n = k + 1$

$WTP: 9 + 14 + 19 + \cdots + (5k + 4) + [5(k + 1) + 4] = \dfrac{5(k + 1)^2 + 13(k + 1)}{2}$

$= \dfrac{5k^2 + 23k + 18}{2}$

$LHS = [9 + 14 + 19 + \cdots + (5k + 4)] + (5k + 9)$

$= \dfrac{5k^2 + 13k}{2} + 5k + 9$, using assumption

$= \dfrac{5k^2 + 13k + 2(5k + 9)}{2}$

$= \dfrac{5k^2 + 23k + 18}{2}$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

g

$2(1!) + 5(2!) + 10(3!) + \cdots + (n^2 + 1)n! = n(n+1)! \quad \text{for all } n \geq 1$

Test $n = 1$

$LHS = 2(1!) = 2 \qquad RHS = 1(1+1)! = 2! = 2$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$2(1!) + 5(2!) + 10(3!) + \cdots + (k^2 + 1)k! = k(k+1)!$

Prove $n = k + 1$

$WTP: 2(1!) + 5(2!) + 10(3!) + \cdots + (k^2 + 1)k! + ((k+1)^2 + 1)(k+1)!$
$= (k+1)(k+2)!$

$$\begin{aligned} LHS &= 2(1!) + 5(2!) + 10(3!) + \cdots + (k^2 + 1)k! + ((k+1)^2 + 1)(k+1)! \\ &= k(k+1)! + ((k+1)^2 + 1)(k+1)!, \text{using assumption} \\ &= (k+1)!\,(k + ((k+1)^2 + 1)) \\ &= (k+1)!\,(k + (k^2 + 2k + 1 + 1)) \\ &= (k+1)!\,(k^2 + 3k + 2) \\ &= (k+1)!\,(k+1)(k+2) \\ &= (k+1)(k+2)(k+1)! \\ &= (k+1)(k+2)! \\ &= RHS \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

h

$5 + 11 + 19 \cdots + (8k - 3) = n(4n+1) \quad \text{for all } n \geq 1$

Test $n = 1$

$LHS = 5 \qquad RHS = 1(4(1) + 1) = 5$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$5 + 11 + 19 + \cdots + (8k - 3) = k(4k+1)$

Prove $n = k + 1$

$WTP: 5 + 11 + 19 + \cdots + (8k - 3) + (8(k+1) - 3) = (k+1)(4(k+1) + 1)$
$= (k+1)(4k+5)$

$$\begin{aligned} LHS &= [5 + 11 + 19 + \cdots + (8k - 3)] + (8k + 5) \\ &= k(4k+1) + (8k+5), \text{using assumption} \\ &= 4k^2 + 9k + 5 \\ &= (k+1)(4k+5) \\ &= RHS \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

i

$$\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\cdots+\frac{n}{(n+1)!}=1-\frac{1}{(n+1)!} \quad \text{for all } n \geq 1$$

Test $n = 1$

$$LHS = \frac{1}{2!}=\frac{1}{2} \qquad RHS = 1-\left(\frac{1}{(1+1)!}\right)=1-\frac{1}{2!}=1-\frac{1}{2}=\frac{1}{2}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$\left(\frac{1}{2!}\right)+\left(\frac{2}{3!}\right)+\left(\frac{3}{4!}\right)+\cdots+\left(\frac{k}{(k+1)!}\right)=1-\left(\frac{1}{(k+1)!}\right)$$

Prove $n = k + 1$

$$WTP: \left(\frac{1}{2!}\right)+\left(\frac{2}{3!}\right)+\left(\frac{3}{4!}\right)+\cdots+\left(\frac{k}{(k+1)!}\right)+\left(\frac{k+1}{(k+2)!}\right)=1-\left(\frac{1}{(k+2)!}\right)$$

$$LHS = \left(\frac{1}{2!}\right)+\left(\frac{2}{3!}\right)+\left(\frac{3}{4!}\right)+\cdots+\left(\frac{k}{(k+1)!}\right)+\left(\frac{k+1}{(k+2)!}\right)$$

$$= 1-\left(\frac{1}{(k+1)!}\right)+\frac{k+2}{(k+2)!}, \text{using assumption}$$

$$= 1-\frac{k+2}{(k+2)(k+1)!}+\left(\frac{k+1}{(k+2)!}\right)$$

$$= 1-\frac{k+2}{(k+2)!}+\left(\frac{k+1}{(k+2)!}\right)$$

$$= 1-\left(\frac{1}{(k+2)!}\right)$$

$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

j

$1(3) + 2(4) + 3(5) + \cdots + n(n+2) = \frac{1}{6}n(n+1)(2n+7) \quad \text{for all } n \geq 1$

Test $n = 1$

$LHS = 1 \times 3 = 3 \qquad RHS = \frac{1}{6}(1)(1+1)(2+7) = \frac{1}{6} \times 1 \times 2 \times 9 = 3$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$1(3) + 2(4) + 3(5) + \cdots + k(k+2) = \frac{1}{6}k(k+1)(2k+7)$

Prove $n = k + 1$

$WTP: 1(3) + 2(4) + 3(5) + \cdots + k(k+2) + (k+1)\big((k+1)+2\big)$

$= \frac{1}{6}(k+1)\big((k+1)+1\big)(2(k+1)+7)$

$= \frac{1}{6}(k+1)(k+2)(2k+9)$

$LHS = 1(3) + 2(4) + 3(5) + \cdots + k(k+2) + (k+1)\big((k+1)+2\big)$

$= \frac{1}{6}k(k+1)(2k+7) + (k+1)\big((k+1)+2\big)$, using assumption

$= \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3)$

$= (k+1)\left(\frac{1}{6}k(2k+7) + k + 3\right)$

$= \frac{1}{6}(k+1)(k(2k+7) + 6k + 18)$

$= \frac{1}{6}(k+1)(2k^2 + 7k + 6k + 18)$

$= \frac{1}{6}(k+1)(2k^2 + 13k + 18)$

$= \frac{1}{6}(k+1)(k+2)(2k+9)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

k

$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \quad \text{for all } n \geq 1$

Test $n = 1$

$LHS = 1 \qquad RHS = \left(\frac{1}{3}\right)(1)(2(1)-1)(2(1)+1) = \frac{1}{3} \times 1 \times 1 \times 3 = 1$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$

Prove $n = k + 1$

$WTP: 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2$

$= \frac{1}{3}(k+1)(2(k+1)-1)(2(k+1)+1)$

$= \frac{1}{3}(k+1)(2k+1)(2k+3)$

$LHS = 1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 + (2(k+1)-1)^2$

$= \frac{1}{3}k(2k-1)(2k+1) + (2(k+1)-1)^2$, using assumption

$= \frac{1}{3}k(2k-1)(2k+1) + (2k+2-1)^2$

$= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2$

$= (2k+1)\left(\frac{1}{3}k(2k-1) + (2k+1)\right)$

$= \frac{1}{3}(2k+1)\left(k(2k-1) + 3(2k+1)\right)$

$= \frac{1}{3}(2k+1)(2k^2 - k + 6k + 3)$

$= \frac{1}{3}(2k+1)(2k^2 + 5k + 3)$

$= \frac{1}{3}(2k+1)(k+1)(2k+3)$

$= \frac{1}{3}(k+1)(2k+1)(2k+3)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

I

$9 + 27 + 81 + \ldots + 3^n = \dfrac{9(3^{n-1} - 1)}{2}$ for all $n \geq 2$ ← The base case is $n = 2$

Test $n = 2$

$LHS = 3^2 = 9$ $\quad RHS = \dfrac{9(3^{2-1} - 1)}{2} = 9$

∴ True for $n = 2$

Assume true for $n = k$

$9 + 27 + 81 + \cdots + 3^k = \dfrac{9(3^{k-1} - 1)}{2}$

Prove $n = k + 1$

$WTP: 9 + 27 + 81 + \cdots + 3^k + 3^{k+1} = \dfrac{9(3^k - 1)}{2}$

$$LHS = [9 + 27 + 81 + \cdots + 3^k] + 3^{k+1}$$
$$= \frac{9(3^{k-1} - 1)}{2} + 3^{k+1}, \text{ using assumption}$$
$$= \frac{9(3^{k-1} - 1) + 2 \times 3^{k+1}}{2}$$
$$= \frac{3 \times 3^k - 9 + 6 \times 3^k}{2}$$
$$= \frac{9 \times 3^k - 9}{2}$$
$$= \frac{9(3^k - 1)}{2}$$
$$= RHS$$

Since true for $n = 2$, by mathematical induction, true for all integers $n \geq 2$.

m

$$1(2^2) + 2(3^2) + 3(4^2) + \cdots + n(n+1)^2 = \frac{1}{12}n(n+1)(n+2)(3n+5) \quad \text{for all } n \geq 1$$

Test $n = 1$

$LHS = 1 \times 2^2 = 4 \quad RHS = \left(\frac{1}{12}\right)(1)(1+1)(1+2)(3+5) = 4$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \cdots + k(k+1)^2 = \frac{1}{12}k(k+1)(k+2)(3k+5)$$

Prove $n = k + 1$

$$WTP: 1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \cdots + k(k+1)^2 + (k+1)\big((k+1)+1\big)^2$$

$$= \frac{1}{12}(k+1)\big((k+1)+1\big)\big((k+1)+2\big)(3(k+1)+5)$$

$$= \frac{1}{12}(k+1)(k+2)(k+3)(3k+8)$$

$$LHS = 1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \cdots + k(k+1)^2 + (k+1)\big((k+1)+1\big)^2$$

$$= \frac{1}{12}k(k+1)(k+2)(3k+5) + (k+1)\big((k+1)+1\big)^2, \text{using assumption}$$

$$= \frac{1}{12}k(k+1)(k+2)(3k+5) + (k+1)(k+2)^2$$

$$= (k+1)(k+2)\left(\frac{1}{12}k(3k+5) + (k+2)\right)$$

$$= \frac{1}{12}(k+1)(k+2)\big(k(3k+5) + 12(k+2)\big)$$

$$= \frac{1}{12}(k+1)(k+2)(3k^2 + 5k + 12k + 24)$$

$$= \frac{1}{12}(k+1)(k+2)(3k^2 + 17k + 24)$$

$$= \frac{1}{12}(k+1)(k+2)(k+3)(3k+8)$$

$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

n

$$-4 + 8 - 16 + \ldots - 4(-2)^{n-1} = \frac{4[(-2)^n - 1]}{3} \quad \text{for all } n \geq 1$$

Test $n = 1$

$$LHS = -4 \qquad RHS = \frac{4((-2)^1 - 1)}{3} = -4$$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$-4 + 8 - 16 + \cdots - 4(-2)^{k-1} = \frac{4\left((-2)^k - 1\right)}{3}$$

Prove $n = k + 1$

$$WTP: -4 + 8 - 16 + \cdots - 4(-2)^{k-1} - 4(-2)^k = \frac{4((-2)^{k+1} - 1)}{3}$$

$$LHS = [-4 + 8 - 16 + \cdots - 4(-2)^{k-1}] - 4(-2)^k$$

$$= \frac{4\left((-2)^k - 1\right)}{3} - 4(-2)^k, \text{using assumption}$$

$$= \frac{4[(-2)^k - 1] - 3 \times -4(-2)^k}{3}$$

$$= \frac{4^2(-2)^k - 4}{3}$$

$$= \frac{4(-2)^{k+1} - 4}{3}$$

$$= \frac{4[(-2)^{k+1} - 1]}{3}$$

$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

o

$1 + 27 + 125 + \ldots + (2n - 1)^3 = n^2(2n^2 - 1)$ for all $n \geq 1$

Test $n = 1$

$LHS = 1 \qquad RHS = 1^2(2 \times 1^2 - 1) = 1$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$1 + 27 + 125 + \cdots + (2k - 1)^3 = k^2(2k^2 - 1)$

Prove $n = k + 1$

$WTP: 1 + 27 + 125 + \cdots + (2k - 1)^3 + (2(k + 1) - 1)^3$

$= (k + 1)^2(2(k + 1)^2 - 1)$

$= 2k^4 + 8k^3 + 11k^2 + 6k + 1$

$LHS = [1 + 27 + 125 + \cdots + (2k - 1)^3] + (2k + 1)^3$

$= k^2(2k^2 - 1) + (2k + 1)^3$, using assumption

$= 2k^4 - k^2 + 8k^3 + 12k^2 + 6k + 1$

$= 2k^4 + 8k^3 + 11k^2 + 6k + 1$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

Proofs of Divisibility

ⓘ Proof by Mathematical Induction

We can use mathematical induction to prove statements where n is a positive whole number.

1. Test the base case, usually $n = 1$
2. Assume $n = k$ is true.
3. Prove $n = k + 1$ is true.
4. State proof by induction.

Example Prove divisibility by mathematical induction.

Prove by mathematical induction:

$$4^n - 1 \text{ is divisible by } 3 \text{ for } n = 1,2,3 \ldots$$

1. Test $n = 1$

$LHS = 4 - 1 = 3$

$\therefore$ True for $n = 1$

2. Assume true for $n = k$

$4^k - 1 = 3p$, where p is an integer.

3. Prove $n = k + 1$

WTP: $4^{k+1} - 1 = 3q$, where q is an integer.

$$\begin{aligned} LHS &= 4^{k+1} - 1 \\ &= 4 \times \underline{4^k} - 1 \\ &= 4 \times \underline{(3p + 1)} - 1, \text{ using assumption} \\ &= 12q + 4 - 1 \\ &= 12q + 3 \\ &= 3(4p + 1), \text{ since } p \text{ is an integer, } 4p + 1 \text{ is also an integer.} \\ &= 3q, \text{ where } q \text{ is an integer} \end{aligned}$$

4. State proof by mathematical induction

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

Practice

Prove by mathematical induction:

$$n(n+1) \text{ is divisible by } 2 \text{ for } n = 1,2,3 \ldots$$

1. Test $n = 1$

2. Assume $n = k$

3. Prove $n = k + 1$

4. State proof by mathematical induction

Prove by mathematical induction:

$$4^n + 14 \text{ is a multiple of } 6 \text{ for } n = 1,2,3 \ldots$$

1 Prove by mathematical induction.

a $7^n - 1$ is divisible by 6 for all positive integers n

Test $n = 1$
$LHS = 7^1 - 1 = 6$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$7^k - 1 = 6p$, where p is an integer.

Prove $n = k + 1$
$WTP: 7^{k+1} - 1 = 6q$, where q is an integer.

$$\begin{aligned} LHS &= 7^{k+1} - 1 \\ &= 7 \times 7^k - 1 \\ &= 7(6p + 1) - 1, \text{ using assumption.} \\ &= 42p + 7 - 1 \\ &= 42p + 6 \\ &= 6(7p + 1), \text{ since } p \text{ is an integer, } 7p + 1 \text{ is also an integer.} \\ &= 6q, \text{ where } q \text{ is an integer.} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

b $5^n - 1$ is divisible by 4 for all integers $n \geq 1$

Test $n = 1$
$LHS = 5^1 - 1 = 4$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$5^k - 1 = 4p$, where p is an integer.

Prove $n = k + 1$
$WTP: 5^{k+1} - 1 = 4q$, where q is an integer.

$$\begin{aligned} LHS &= 5^{k+1} - 1 \\ &= 5 \times 5^k - 1 \\ &= 5(4p + 1) - 1, \text{ using assumption} \\ &= 20p + 5 - 1 \\ &= 20p + 4 \\ &= 4(5p + 1), \text{ since } p \text{ is an integer, } 5p + 1 \text{ is also an integer.} \\ &= 4q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

c $3^{2n} - 1$ is divisible by 8 for all integers $n \geq 1$

Test $n = 1$
$LHS = 3^2 - 1 = 9 - 1 = 8$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$3^{2k} - 1 = 8p$, where p is an integer.

Prove $n = k + 1$
$WTP: 3^{2(k+1)} - 1 = 8q$, where q is an integer.
$LHS = 3^{2k+2} - 1$
$= 3^2 \times 3^{2k} - 1$
$= 9 \times 3^{2k} - 1$
$= 9(8p + 1) - 1$, using assumption
$= 72p + 9 - 1$
$= 72p + 8$
$= 8(9p + 1)$, since p is an integer, $9p + 1$ is also an integer.
$= 8q$, where q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

---

d Fill in the gaps.
Prove $3^{2n} - 1$ is divisible by 8 for all integers $n \geq 1$

..... $n = 1$:

Test

$LHS = 3^2 - 1 = 9 - 1 = 8$
$\therefore$ .... $n = 1$

True for

.... $n = k$:

Assume true for

$3^{2k} - 1 = 8p$, where ....

k is an integer

.... $n = k + 1$:

Prove true for

.....: $3^{2(k+1)} - 1 = 8q$, ...

WTP: where q is an integer

..... $= 3^{2k+2} - 1$

LHS

$= 3^2 \times 3^{2k} - 1$
$= 9 \times 3^{2k} - 1$
$= 9(8p + 1) - 1$, ...

Using asummption

$= 72p + 9 - 1$
$= 72p + 8$
$= 8(9p + 1)$, since p ....., ... is also .....

is an integer, $9p + 1$ is also an integer

$= 8q$, ...

Where q is an integer

.....

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

e $3^{2n} + 7$ is divisible by 8 for all integers $n \geq 1$

Test $n = 1$
$LHS = 3^2 + 7 = 9 + 7 = 16$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$3^{2k} + 7 = 8p$, where p is an integer.

Prove $n = k + 1$
$WTP: 3^{2(k+1)} + 7 = 8q$, where q is an integer.

$$\begin{aligned} LHS &= 3^{2k+2} + 7 \\ &= 9 \times 3^{2k} + 7 \\ &= 9(8p - 7) + 7, \text{ using assumption} \\ &= 72p - 63 + 7 \\ &= 72p - 56 \\ &= 8(9p - 7), \text{ since } p \text{ is an integer, } 9p - 7 \text{ is also an integer.} \\ &= 8q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

---

f $5^{2n} - 1$ is divisible by 24 for all integers $n \geq 1$

Test $n = 1$
$LHS = 5^2 - 1 = 25 - 1 = 24$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$5^{2k} - 1 = 24p$, where p is an integer.

Prove $n = k + 1$
$WTP: 5^{2(k+1)} - 1 = 24q$, where q is an integer.

$$\begin{aligned} LHS &= 5^{2k+2} - 1 \\ &= 5^2 \times 5^{2k} - 1 \\ &= 25 \times 5^{2k} - 1 \\ &= 25(24p + 1) - 1, \text{ using assumption} \\ &= 600p + 25 - 1 \\ &= 600p + 24 \\ &= 24(25p + 1), \text{ since } p \text{ is an integer, } 25p + 1 \text{ is also an integer.} \\ &= 24q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

g $n^3 + 2n$ is divisible by 3 for all integers $n \geq 0$

Test $n = 0$
$LHS = 0^3 + 2(0) = 0$
$\therefore$ True for $n = 0$

Assume true for $n = k$
$k^3 + 2k = 3p$, where p is a an integer.

Prove $n = k + 1$
$WTP: (k+1)^3 + 2(k+1) = 3q$, where q is an integer.
$LHS = (k+1)^3 + 2(k+1)$
$= k^3 + 3k^2 + 3k + 1 + 2k + 2$
$= k^3 + 2k + 3k^2 + 3k + 3$
$= (k^3 + 2k) + 3k^2 + 3k + 3$
$= 3p + 3k^2 + 3k + 3$, using assumption
$= 3(p + k^2 + k + 1)$, since p and k are integers,
$p + k^2 + k + 1$ is also an integer.
$= 3q$, where q is an integer

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

h $9^n + 3$ is divisible by 6 for all integers $n \geq 1$

Test $n = 1$
$LHS = 9^1 + 3 = 9 + 3 = 12 = 6(2)$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$9^k + 3 = 6p$, where p is an integer.

Prove $n = k + 1$
$WTP: 9^{k+1} + 3 = 6q$, where q is an integer.

$LHS = 9^{k+1} + 3$
$= 9 \times 9^k + 3$
$= 9(6p - 3) + 3$, using assumption
$= 54p - 27 + 3$
$= 54p - 24$
$= 6(9p - 4)$, since p is an integer, $9p - 4$ is also an integer.
$= 6q$, where q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

i Consider the two proofs for "$5^n + 3^n$ is even for all positive integers n", shown below.

Test $n = 1$
$LHS = 5^1 + 3^1 = 5 + 3 = 8 = 2(4)$
$\therefore$ True for $n = 1$

Assume true for $n = k$
$5^k + 3^k = 2p$, where p is an integer.

Prove $n = k + 1$
WTP: $5^{k+1} + 3^{k+1} = 2q$, where q is an integer.

Method 1:
$$\begin{aligned} LHS &= 5^{k+1} + 3^{k+1} \\ &= 5 \cdot 5^k + 3 \cdot 3^k \\ &= 2 \cdot 5^k + 3 \cdot 5^k + 3 \cdot 3^k \\ &= 2 \cdot 5^k + 3(5^k + 3^k) \\ &= 2 \cdot 5^k + 3(2p), \text{ using assumption} \\ &= 2 \cdot 5^k + 6p \\ &= 2(5^k + 3p), \text{ since } 5^k \text{ and } p \text{ are integers, } 5^k + 3p \text{ is also an integer.} \\ &= 2q, \text{ where } q \text{ is an integer.} \end{aligned}$$

Method 2:
$$\begin{aligned} LHS &= 5^{k+1} + 3^{k+1} \\ &= 5 \cdot 5^k + 3 \cdot 3^k \\ &= 5(2p - 3^k) + 3 \cdot 3^k, \text{ using assumption} \\ &= 10p - 5 \cdot 3^k + 3 \cdot 3^k \\ &= 10p - 2 \cdot 3^k \\ &= 2(5p - 3^k) \text{ since } p \text{ and } 3^k \text{ are positive integers, } 5p - 3^k \text{ is also an integer.} \\ &= 2q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

What is the difference between the two methods, and which do you prefer?

Method 1 splits the 5^{k+1} and factors out the common term $(5^k + 3^k)$ explicitly, then substitutes the inductive hypothesis directly: $5^k + 3^k = 2p$.

Method 2 requires rearranging the inductive hypothesis; 5^k is replaced with $(2p - 3^k)$.

Method 2 is easier to apply, but method 1 may be required for harder proofs in Extension 2.

j $7^n + 3^n$ is divisible by 10 for all odd positive integers $n \geq 1$
Hint: assume true for $n = 2k - 1$

Test $n = 1$
$LHS = 7^1 + 3^1 = 10$
$\therefore$ True for $n = 1$

Assume true for $n = 2k - 1$
$7^{2k-1} + 3^{2k-1} = 10p$, where p is an integer.

Prove $n = 2k + 1$
$WTP: 7^{2k+1} + 3^{2k+1} = 10q$, where q is an integer.

$$\begin{aligned} LHS &= 7^{2k+1} + 3^{2k+1} \\ &= 7^2 \times 7^{2k-1} + 3^2 \times 3^{2k-1} \\ &= 49 \times 7^{2k-1} + 9 \times 3^{2k-1} \\ &= 49(10p - 3^{2k-1}) + 9 \times 3^{2k-1} \\ &= 490p - 49 \times 3^{2k-1} + 9 \times 3^{2k-1} \\ &= 490p - 40 \times 3^{2k-1} \\ &= 10(49p - 4 \times 3^{2k-1}), \text{ since } p \text{ and } 3^{2k-1} \text{ are integers,} \\ &\qquad 49p - 4 \times 3^{2k-1} \text{ is also an integer.} \\ &= 10q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all odd integers $n \geq 1$.

---

k $5^n + 2 \times 11^n$ is divisible by 3 for all integers $n \geq 0$

Test $n = 0$
$LHS = 5^0 + 2 \times 11^0 = 1 + 2 = 3$
$\therefore$ True for $n = 0$

Assume true for $n = k$
$5^k + 2 \times 11^k = 3p$, where p is an integer.

Prove $n = k + 1$
$WTP: 5^{k+1} + 2 \times 11^{k+1} = 3q$, where q is an integer.

$$\begin{aligned} LHS &= 5^{k+1} + 2 \times 11^{k+1} \\ &= 5 \times 5^k + 2 \times 11 \times 11^k \\ &= 5 \times 5^k + 22 \times 11^k \\ &= 5(3p - 2 \times 11^k) + 22 \times 11^k \\ &= 15p - 10 \times 11^k + 22 \times 11^k \\ &= 15p + 12 \times 11^k \\ &= 3(5p + 4 \times 11^k), \text{ since } p \text{ and } 11^k \text{ are integers, } 5p + 4 \times 11^k \text{ is also an integer.} \\ &= 3q, \text{ where } q \text{ is an integer} \end{aligned}$$

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

l $8^n - 7n + 6$ is divisible by 7 for all integers $n \geq 0$

Test $n = 0$
$LHS = 8^0 - 7(0) + 6 = 1 + 6 = 7$
$\therefore$ True for $n = 0$

Assume true for $n = k$
$8^k - 7k + 6 = 7p$, where p is an integer.

Prove $n = k + 1$
$WTP: 8^{k+1} - 7(k + 1) + 6 = 7q$, where q is an integer.
$LHS = 8^{k+1} - 7(k + 1) + 6$
$= 8 \times 8^k - 7k - 7 + 6$
$= 8 \times 8^k - 7k - 1$
$= 8(7p + 7k - 6) - 7k - 1$
$= 56p + 56k - 48 - 7k - 1$
$= 56p + 49k - 49$
$= 7(8p + 7k - 7)$, since p and k are integers,
$8p + 7k - 7$ is also an integer.
$= 7q$, where q is an integer

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

m Prove that the sum of consecutive odd positive integers is divisible by 4.
In other words, $1 + 3 + 5 + \ldots + (2n - 1)$ is divisible by 4 for $n \geq 2$ and even.
Hint: assume true for $n = 2k$

Test $n = 2$
$LHS = 1 + 3 = 4$
$\therefore$ True for $n = 2$

Assume true for $n = 2k$
$1 + 3 + 5 + \ldots + (4k - 1) = 4p$, where p is an integer.

Prove $n = 2k + 2$
$WTP: 1 + 3 + 5 + \ldots + (4k - 1) + (4k + 1) + (4k + 3) = 4q$, where q is an integer.
$LHS = 1 + 3 + 5 + \ldots + (4k - 1) + (4k + 1) + (4k + 3)$
$= 4p + (4k + 1) + (4k + 3)$, using assumption
$= 4p + 8k + 4$
$= 4(p + 2k + 1)$, since p and k are integers,
$p + 2k + 1$ is also an integer.
$= 4q$, where q is an integer

Since true for $n = 2$, by mathematical induction, true for all even integers $n \geq 2$.

n $9(9^n - 1) - 8n$ is divisible by 64 for all integers $n \geq 0$

Test $n = 0$

$LHS = 9(9^0 - 1) - 8(0) = 0 = 64(0)$

$\therefore$ True for $n = 0$

Assume true for $n = k$

$9(9^k - 1) - 8k = 64p$, where p is a non-negative integer.

Prove $n = k + 1$

$WTP: 9(9^{k+1} - 1) - 8(k + 1) = 64q$, where q is a non-negative integer.

$LHS = 81 \times 9^k - 9 - 8k - 8$

$= 81 \times 9^k - 8k - 17$ From assumption: $9(9^k - 1) - 8k = 64p$

So: $9 \times 9^k - 9 = 64p + 8k$

$\therefore 9 \times 9^k = 64p + 8k + 9$

$= 9 \times 9 \times 9^k - 8k - 17$

$= 9(64p + 8k + 9) - 8k - 17$

$= 576p + 72k + 81 - 8k - 17$

$= 576p + 64k + 64$

$= 64(9p + k + 1)$, since p and k are non-negative integers,

$9p + k + 1$ is also a non-negative integer.

$= 64q$, where q is a non-negative integer

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

o $3^{3n} + 2^{n+2}$ is divisible by 5 for all integers $n \geq 0$

Test $n = 0$

$LHS = 3^0 + 2^2 = 1 + 4 = 5$

$\therefore$ True for $n = 0$

Assume true for $n = k$

$3^{3k} + 2^{k+2} = 5p$, where p is a non-negative integer.

Prove $n = k + 1$

$WTP: 3^{3(k+1)} + 2^{(k+1)+2} = 5q$, where q is a non-negative integer.

$LHS = 3^{3k+3} + 2^{k+3}$

$= 3^3 \times 3^{3k} + 2 \times 2^{k+2}$

$= 27 \times 3^{3k} + 2 \times 2^{k+2}$

$= 27(5p - 2^{k+2}) + 2 \times 2^{k+2}$, using assumption.

$= 135p - 27 \times 2^{k+2} + 2 \times 2^{k+2}$

$= 135p - 25 \times 2^{k+2}$

$= 5(27p - 5 \times 2^{k+2})$, since p and 2^{k+2} are integers, $27p - 5 \times 2^{k+2}$ is also an integer.

$= 5q$, where q is an integer

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

p $11^{n+2} + 12^{2n+1}$ is divisible by 133 for all integers $n \geq 0$

Test $n = 0$

$LHS = 11^2 + 12^1 = 121 + 12 = 133$

$\therefore$ True for $n = 0$

Assume true for $n = k$

$11^{k+2} + 12^{2k+1} = 133p$, where p is a non-negative integer.

Prove $n = k + 1$

$WTP: 11^{(k+1)+2} + 12^{2(k+1)+1} = 133q$, where q is an integer.

$$\begin{aligned} LHS &= 11^{k+3} + 12^{2k+3} \\ &= 11 \times 11^{k+2} + 12^2 \times 12^{2k+1} \\ &= 11 \times 11^{k+2} + 144 \times 12^{2k+1} \\ &= 11(133p - 12^{2k+1}) + 144 \times 12^{2k+1}, \text{ using assumption} \\ &= 1463p - 11 \times 12^{2k+1} + 144 \times 12^{2k+1} \\ &= 1463p + 133 \times 12^{2k+1} \\ &= 133(11p + 12^{2k+1}), \text{ since } p \text{ and } 12^{2k+1} \text{ are integers,} \\ &\qquad 11p + 12^{2k+1} \text{ is also an integer.} \\ &= 133q, \text{ where } q \text{ is an integer.} \end{aligned}$$

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

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q Prove by mathematical induction that $x - 1$ is a factor of $x^n - 1$ for all integers $n \geq 1$

Test $n = 1$

$LHS = x^1 - 1 = x - 1$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$x^k - 1 = (x - 1)p(x)$, where $p(x)$ is a polynomial.

Prove $n = k + 1$

$WTP: x^{k+1} - 1 = (x - 1)q(x)$, where $q(x)$ is a polynomial.

$$\begin{aligned} LHS &= x^{k+1} - 1 \\ &= x \times x^k - 1 \\ &= x[(x-1)p(x) + 1] - 1, \text{ using assumption} \\ &= x(x-1)p(x) + x - 1 \\ &= (x - 1)(xp(x) + 1) \\ &= (x - 1)q(x), \text{ where } q(x) = xp(x) + 1 \text{ is a polynomial} \end{aligned}$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

Exam Questions

1 2014 HSC Advanced Band 3

Use mathematical induction to prove that $2^n + (-1)^{n+1}$ is divisible by 3 for all integers $n \geq 1$.

Test n = 1

$LHS = 2^1 + (-1)^2 = 2 + 1 = 3$

$\therefore$ True for n = 1

Assume true for $n = k$

$2^k + (-1)^{k+1} = 3P$, where P is an integer.

Rearranging: $2^k = 3P - (-1)^{k+1}$

Prove $n = k + 1$

WTP: $2^{k+1} + (-1)^{k+2}$ is divisible by 3.

$LHS = 2^{k+1} + (-1)^{k+1+1}$

$= 2 \times 2^k + (-1)^{k+2}$

$= 2(3P - (-1)^{k+1}) + (-1)^{k+2}$, from assumption

$= 6P - 2(-1)^{k+1} - 1(-1)^{k+1}$

$= 6P - 3(-1)^{k+1}$

$= 3(2P - (-1)^{k+1})$

$= 3Q$, where Q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

2 2023 HSC Advanced Band 3

Use mathematical induction to prove that
$(1 \times 2) + (2 \times 2^2) + (3 \times 2^3) + \ldots + (n \times 2^n) = 2 + (n-1)2^{n+1}$ for all integers $n \geq 1$.

Test $n = 1$

$LHS = 1 \times 2 = 2$

$RHS = 2 + (1-1)2^2 = 2$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$(1 \times 2) + (2 \times 2^2) + \ldots + (k \times 2^k) = 2 + (k-1)2^{k+1}$

Prove $n = k + 1$

$WTP: (1 \times 2) + (2 \times 2^2) + \ldots + (k \times 2^k) + (k+1)2^{k+1} = 2 + k \times 2^{k+2}$

$LHS = 2 + (k-1)2^{k+1} + (k+1)2^{k+1}$

$= 2 + 2^{(k+1)}(k - 1 + k + 1)$

$= 2 + 2^{(k+1)}(2k)$

$= 2 + k \times 2^{k+2}$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

3 2024 HSC Advanced Band 3

Use mathematical induction to prove that $2^{3n} + 13$ is divisible by 7 for all integers $n \geq 1$.

Test $n = 1$

$LHS = 2^{3\times1} + 13 = 21 = 3 \times 7$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$2^{3k} + 13 = 7P$, where P is an integer.

Rearranging: $2^{3k} = 7P - 13$

Prove $n = k + 1$

$WTP: 2^{3(k+1)} + 13$ is divisible by 7.

$LHS = 2^{3k+3} + 13$

$= 2^3 \times 2^{3k} + 13$

$= 8(2^{3k}) + 13$

$= 8(7P - 13) + 13, \text{using assumption}$

$= 56P - 8 \times 13 + 13$

$= 7(8P - 13)$

$= 7Q$, where Q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

4 2016 HSC Advanced Band 4

a Show that $4n^3 + 18n^2 + 23n + 9$ can be written as $(n + 1)(4n^2 + 14n + 9)$.

$RHS = (n + 1)(4n^2 + 14n + 9)$

$= 4n^3 + 14n^2 + 9n + 4n^2 + 14n + 9$

$= 4n^3 + 18n^2 + 23n + 9$

$= LHS$

b Using the result in part (a), or otherwise, prove by mathematical induction that, for $n \geq 1$,

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \ldots + (2n - 1)(2n + 1) = \frac{1}{3}n(4n^2 + 6n - 1).$$

Test $n = 1$

$LHS = (1)(3) = 3$

$RHS = \left(\frac{1}{3}\right)(1)(4 + 6 - 1) = 3$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$1 \times 3 + 3 \times 5 + \ldots + (2k - 1)(2k + 1) = \frac{1}{3}k(4k^2 + 6k - 1)$

Prove $n = k + 1$

$WTP: 1 \times 3 + \ldots + (2k + 1)(2k + 3) = \frac{1}{3}(k + 1)[4(k + 1)^2 + 6(k + 1) - 1]$

$= \frac{1}{3}(k + 1)(4k^2 + 14k + 9)$

$= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9)$

$LHS = 1 \times 3 + \ldots + (2k - 1)(2k + 1) + (2k + 1)(2k + 3)$

$= \frac{1}{3}k(4k^2 + 6k - 1) + (2k + 1)(2k + 3)$

$= \frac{1}{3}(4k^3 + 6k^2 - k) + (4k^2 + 8k + 3)$

$= \frac{1}{3}(4k^3 + 6k^2 - k + 12k^2 + 24k + 9)$

$= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9)$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

5 2007 HSC Advanced Band 4

Use mathematical induction to prove that $7^{2n-1} + 5$ is divisible by 12, for all integers $n \geq 1$.

Test $n = 1$

$LHS = 7^{2-1} + 5 = 7 + 5 = 12$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$7^{2k-1} + 5 = 12N$, where N is an integer.

Rearranging: $7^{2k-1} = 12N - 5$

Prove $n = k + 1$

WTP: $7^{2(k+1)-1} + 5$ is divisible by 12.

$LHS = 7^{2k+1} + 5$

$= 7^2 \times 7^{2k-1} + 5$

$= 49(12N - 5) + 5$, using assumption

$= 49 \times 12N - 245 + 5$

$= 49 \times 12N - 240$

$= 12(49N - 20)$

$= 12Q$, where Q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

6 2017 HSC Advanced Band 4

Prove by mathematical induction that $8^{2n+1} + 6^{2n-1}$ is divisible by 7, for any integer $n \geq 1$.

Test $n = 1$

$LHS = 8^3 + 6^1 = 518 = 74 \times 7$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$8^{2k+1} + 6^{2k-1} = 7P$, where P is an integer.

Rearranging: $8^{2k+1} = 7P - 6^{2k-1}$

Prove $n = k + 1$

WTP: $8^{2(k+1)+1} + 6^{2(k+1)-1}$ is divisible by 7.

$LHS = 8^{2k+3} + 6^{2k-1}$

$= 64 \times 8^{2k+1} + 36 \times 6^{2k-1}$

$= 64(7P - 6^{2k-1}) + 36 \times 6^{2k-1}$, using assumption

$= 64 \times 7P - 64 \times 6^{2k-1} + 36 \times 6^{2k-1}$

$= 64 \times 7P - 28 \times 6^{2k-1}$

$= 7(64P - 4 \times 6^{2k-1})$

$= 7Q$, where Q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

7 2021 HSC Advanced Band 4

Use mathematical induction to prove that

$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \ldots + \frac{1}{n(n+1)(n+2)} = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$$

for all integers $n \geq 1$.

Test $n = 1$

$$LHS = \frac{1}{1 \times 2 \times 3} = \frac{1}{6}$$

$$RHS = \frac{1}{4} - \frac{1}{2 \times 2 \times 3} = \frac{1}{4} - \frac{1}{12} = \frac{1}{6}$$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$\frac{1}{1 \times 2 \times 3} + \ldots + \frac{1}{k(k+1)(k+2)} = \frac{1}{4} - \frac{1}{2(k+1)(k+2)}$$

Prove $n = k + 1$

$$WTP: \frac{1}{1 \times 2 \times 3} + \ldots + \frac{1}{(k+1)(k+2)(k+3)} = \frac{1}{4} - \frac{1}{2(k+2)(k+3)}$$

$$LHS = \frac{1}{1 \times 2 \times 3} + \ldots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{1}{4} - \frac{1}{2(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{1}{4} - \frac{k+3-2}{2(k+1)(k+2)(k+3)}$$

$$= \frac{1}{4} - \frac{k+1}{2(k+1)(k+2)(k+3)}$$

$$= \frac{1}{4} - \frac{1}{2(k+2)(k+3)}$$

$$= RHS$$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

8 2022 HSC Advanced Band 4

Use mathematical induction to prove that $15^n + 6^{2n+1}$ is divisible by 7 for all integers $n \geq 0$.

Test $n = 0$

$LHS = 15^0 + 6^1 = 7$

$\therefore$ True for $n = 0$

Assume true for $n = k$

$15^k + 6^{2k+1} = 7P$, where P is an integer.

Rearranging: $6^{2k+1} = 7P - 15^k$

Prove $n = k + 1$

WTP: $15^{k+1} + 6^{2k+3}$ is divisible by 7.

$LHS = 15^{k+1} + 6^{2k+3}$

$= 15 \times 15^k + 6^2 \times 6^{2k+1}$

$= 15 \times 15\hat{}k + 36(7P - 15\hat{}k)$, using assumption

$= 15 \times 15^k + 36 \times 7P - 36 \times 15^k$

$= 36 \times 7P - 21 \times 15^k$

$= 7(36P - 3 \times 15^k)$

$= 7Q$, where Q is an integer

Since true for $n = 0$, by mathematical induction, true for all integers $n \geq 0$.

9 HSC Advanced Sample Question Band 5

Prove by mathematical induction that $(3n + 1)7^n - 1$ is divisible by 9 for integers $n \geq 1$.

Hint: prove that $f(x+1) - f(x)$ is a multiple of 9.

Test $n = 1$

$LHS = (3 \times 1 + 1)7^1 - 1 = 27$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$(3k + 1)7^k - 1 = 9P$, where P is an integer.

Prove $n = k + 1$

Let $f(k) = (3k + 1)7^k - 1$

Then $f(k + 1) = (3(k + 1) + 1)7^{k+1} - 1 = (3k + 4)7^{k+1} - 1$

$f(k + 1) - f(k) = (3k + 4)7^{k+1} - 1 - [(3k + 1)7^k - 1]$

$= 3k \times 7^{k+1} + 4 \times 7^{k+1} - 1 - [3k \times 7^k + 7^k - 1]$

$= 21k \times 7^k + 28 \times 7^k - 1 - 3k \times 7^k - 7^k + 1$

$= 18k \times 7^k + 27 \times 7^k$

$= 9(2k \times 7^k + 3 \times 7^k)$

$= 9Q$, where Q is an integer

Since $f(k + 1) - f(k)$ is divisible by 9 and $f(k)$ is divisible by 9,

$\Rightarrow f(k + 1)$ is divisible by 9

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

10 2010 HSC Advanced Band 5

Prove by induction that $47^n + 53 \times 147^{n-1}$ is divisible by 100 for all integers $n \geq 1$.

Test $n = 1$

$LHS = 47^1 + 53 \times 147^{1-1} = 47 + 53 = 100$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$47^k + 53 \times 147^{k-1} = 100P$, where P is an integer.

Rearranging: $47^k = 100P - 53 \times 147^{k-1}$

Prove $n = k + 1$

WTP: $47^{k+1} + 53 \times 147^k$ is divisible by 100.

$LHS = 47^{k+1} + 53 \times 147^k$

$= 47 \times 47^k + 53 \times 147^k$

$= 47[100P - 53 \times 147^{k-1}] + 53 \times 147 \times 147^{k-1}$, using assumption

$= 4700P - 2491 \times 147^{k-1} + 7791 \times 147^{k-1}$

$= 4700P + 5300 \times 147^{k-1}$

$= 100(47P + 53 \times 147^{k-1})$

$= 100Q$, where Q is an integer

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.

11 HSC Advanced Sample Question Band 6

a Show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$.

$\cos(A - B) - \cos(A + B) = \cos A \cos B + \sin A \sin B - (\cos A \cos B - \sin A \sin B)$

$= 2\sin A \cos B$

b Using the result in part (a), or otherwise, prove by mathematical induction that, for $n \geq 1$

$$\sin\theta + \sin(3\theta) + \ldots + \sin\big((2n-1)\theta\big) = \frac{\sin^2(n\theta)}{\sin\theta}$$

Test $n = 1$

$LHS = \sin\theta$

$RHS = \dfrac{\sin^2\theta}{\sin\theta} = \sin\theta$

$\therefore$ True for $n = 1$

Assume true for $n = k$

$$\sin\theta + \sin(3\theta) + \ldots + \sin\big((2k-1)\theta\big) = \frac{\sin^2(k\theta)}{\sin\theta}$$

Prove $n = k + 1$

$$WTP: \sin\theta + \sin(3\theta) + \ldots + \sin\big((2k-1)\theta\big) + \sin\big((2k+1)\theta\big) = \frac{\sin^2\big((k+1)\theta\big)}{\sin\theta}$$

$LHS = \dfrac{\sin^2(k\theta)}{\sin\theta} + \sin\big((2k+1)\theta\big)$, using assumption

$= \dfrac{\sin^2(k\theta) + \sin\theta \times \sin\big((2k+1)\theta\big)}{\sin\theta}$

Using part a:

$\sin\theta \times \sin\big((2k+1)\theta\big) = \dfrac{1}{2}\big[\cos(2k\theta) - \cos\big((2k+2)\theta\big)\big]$

$= \dfrac{1}{2}[\cos(2k\theta) - \cos(2(k+1)\theta)]$

$= \dfrac{1}{2}\Big[1 - 2\sin^2(k\theta) - \big(1 - 2\sin^2\big((k+1)\theta\big)\big)\Big]$

$= \dfrac{1}{2}\Big(2\sin^2\big((k+1)\theta\big) - 2\sin^2(k\theta)\Big)$

$= \sin^2((k+1)\theta) - \sin^2(k\theta)$

Substitute into LHS:

$LHS = \dfrac{\sin^2(k\theta) + \sin^2\big((k+1)\theta\big) - \sin^2(k\theta)}{\sin\theta}$

$= \dfrac{\sin^2\big((k+1)\theta\big)}{\sin\theta}$

$= RHS$

Since true for $n = 1$, by mathematical induction, true for all integers $n \geq 1$.